

Bootstrap — Topics 32-38: Modal, Toast, Tooltip, Popover, Offcanvas, Scrollspy, Placeholder

Study paper format: notes (Roman Urdu + English technical terms), har component ke peeche ka CSS/JS mechanism, + active recall Q&A at the end. Covers: Modal (★ ★ ★), Toast, Tooltip, Popover, Offcanvas (★ ★ ★), Scrollspy, Placeholder.

32. Modal ★ ★ ★

Modal ek **popup dialog box** hai jo page ke upar overlay ho kar dikhta hai — content ko blur/dim background ke sath focus deta hai.

Basic Structure

```
html
```

```

<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal"
  Open Modal
</button>

<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Modal Title</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        Modal body content here.
      </div>
      <div class="modal-footer">
        <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
        <button class="btn btn-primary">Save changes</button>
      </div>
    </div>
  </div>
</div>

```

Open Modal

`data-bs-toggle="modal"` + `data-bs-target="#myModal"` trigger button ko modal se link karta hai. Click hone par Bootstrap JS modal ko open karta hai.

Peeche ki CSS/mechanism:

- Default `.modal` `display: none` hoti hai
- Open hone par JS `display: block` set karta hai aur `.show` class add karta hai — `.fade` + `.show` combination se modal **opacity aur** `transform: translateY()`

transition ke sath smoothly andar/upar se slide ho kar aata hai (thora upar se neeche aane jaisa effect)

- JS ek separate `.modal-backdrop` `<div>` bhi body mein inject karta hai — ye poori screen ko cover karta hai (`position: fixed`), dark semi-transparent (`background-color`), jisse background dim/blurred jaisa lagta hai aur click-outside detect hoti hai
- JS `<body>` par `overflow: hidden` bhi laga deta hai (aur `modal-open` class), taake jab modal open ho, background page scroll na ho sake

Close Modal

`data-bs-dismiss="modal"` kisi bhi button/element par lagane se wo modal close kar deta hai (JS listener). Reverse process: `.show` class remove → fade-out transition → `display: none` → backdrop bhi remove ho jata hai aur body scroll wapas enable ho jata hai.

Large / Small Modal

html

```
<div class="modal-dialog modal-lg">...</div>
<div class="modal-dialog modal-sm">...</div>
```

`.modal-lg` aur `.modal-sm` `.modal-dialog` par `max-width` set karti hain (large ke liye zyada, small ke liye kam) — poori modal box ki width control karti hain.

Centered Modal

html

```
<div class="modal-dialog modal-dialog-centered">...</div>
```

Peeche ki CSS: `.modal-dialog-centered` `.modal-dialog` ko `display: flex` + `align-items: center` + `min-height: calc(100% - margin)` deta hai — isse modal box vertically

screen ke **center** mein aa jata hai (default behavior mein modal top ke qareeb start hoti hai).

33. Toast (Notifications)

Toast ek **choti, temporary notification** hai jo corner mein pop-up hoti hai (jaise "Message sent!") aur khud-ba-khud gayab ho jati hai.

```
html

<div class="toast" role="alert" data-bs-autohide="true" data-bs-delay="3000">
  <div class="toast-header">
    <strong class="me-auto">Notification</strong>
    <small>Just now</small>
    <button type="button" class="btn-close" data-bs-dismiss="toast"></button>
  </div>
  <div class="toast-body">
    Your message has been sent.
  </div>
</div>
```

Multiple toasts ko ek `.toast-container` mein rakha jata hai taake wo consistently ek corner mein stack ho sakein:

```
html

<div class="toast-container position-fixed bottom-0 end-0 p-3">
  <!-- toasts yahan -->
</div>
```

Peeche ki CSS/mechanism:

- Modal ki tarah, `.toast` bhi `.fade` + `.show` opacity transition use karta hai open/close ke liye
 - Farq ye hai ke toast **koi backdrop nahi** banata aur background ko block nahi karta — user baqi page ke sath interact kar sakta hai
 - `data-bs-autohide="true"` aur `data-bs-delay="3000"` JS ko batate hain ke `3000`ms (3 seconds) baad automatically hide ho jaye — JS internally `setTimeout` jaisi logic use karta hai
 - `.toast-container` ke `position-fixed bottom-0 end-0` classes container ko screen ke bottom-right corner par fix kar dete hain, chahe page kitna bhi scroll ho
-

34. Tooltip

Tooltip ek **choti hint/label** hai jo element par hover karne se dikhti hai.

```
html

<button type="button" class="btn btn-secondary" data-bs-toggle="tooltip" title="Hover me">
  Hover me
</button>
```

Tooltips ko manually JavaScript se **initialize** karna zaroori hota hai (dusre components ki tarah automatic nahi):

```
javascript

const tooltipTriggerList = document.querySelectorAll('[data-bs-toggle="tooltip"]')
tooltipTriggerList.forEach(el => new bootstrap.Tooltip(el));
```

Peeche ki CSS/mechanism:

- Bootstrap `title` attribute ko read karta hai aur uski jagah ek custom `.tooltip` `<div>` **dynamically JS se create** karta hai (browser ka default native tooltip use

nahi hota, jo ugly hota hai)

- Positioning ke liye Popper.js library use hoti hai (yehi wo library hai jo `bootstrap.bundle.min.js` mein already included hoti hai) — ye calculate karti hai ke tooltip element ke kis side (top/bottom/left/right) fit hoga bina viewport se bahar jaye
 - `.fade` + `.show` yahan bhi opacity transition ke liye use hoti hain
 - `data-bs-placement="top/bottom/left/right"` se tooltip ki default direction specify ki ja sakti hai
-

35. Popover

Popover, Tooltip jaisa hi hai lekin **zyada content** (title + body) show kar sakta hai, aur usually click par trigger hota hai (hover ki bajaye).

html

```
<button type="button" class="btn btn-secondary" data-bs-toggle="popover" title="Click me">Click me</button>
```

javascript

```
const popoverTriggerList = document.querySelectorAll('[data-bs-toggle="popover"]');  
popoverTriggerList.forEach(el => new bootstrap.Popover(el));
```

Peeche ki CSS/mechanism:

- Popover bhi Popper.js se positioned hota hai, exactly Tooltip jaisa
- Farq structural hai: Popover ke andar `.popover-header` aur `.popover-body` dono hote hain (Tooltip mein sirf ek simple text hota hai)

- Popover mein ek chota **arrow** bhi hota hai jo trigger element ki taraf point karta hai — ye CSS `border` tricks se banaya gaya triangle hota hai, jiski position Popper.js dynamically calculate karta hai

Tooltip vs Popover:

	Tooltip	Popover
Content	Sirf simple text	Title + rich body
Trigger (default)	Hover/focus	Click
Use case	Chota hint	Extra detail/explanation

36. Offcanvas ★ ★ ★ (Sidebar Navigation)

Offcanvas ek **sliding sidebar panel** hai jo screen ke edge se slide-in hota hai — mobile navigation menus ke liye bohat common.

```
html
```

```

<button class="btn btn-primary" data-bs-toggle="offcanvas" data-bs-target="#m
    Open Sidebar
</button>

<div class="offcanvas offcanvas-start" tabindex="-1" id="myOffcanvas">
  <div class="offcanvas-header">
    <h5 class="offcanvas-title">Menu</h5>
    <button type="button" class="btn-close" data-bs-dismiss="offcanvas"><
  </div>
  <div class="offcanvas-body">
    <ul class="navbar-nav">
      <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
      <li class="nav-item"><a class="nav-link" href="#">About</a></li>
    </ul>
  </div>
</div>

```

Directions

```

offcanvas-start  → left se slide-in
offcanvas-end    → right se slide-in
offcanvas-top    → top se slide-in
offcanvas-bottom → bottom se slide-in

```

Peeche ki CSS/mechanism:

- Default offcanvas `position: fixed` hoti hai, screen ke edge (left/right/top/bottom) par, lekin `transform: translateX()` (ya `translateY()` top/bottom ke liye) se **screen se bahar** rakhi jati hai — jaise `offcanvas-start` ke liye `transform: translateX(-100%)` (poori tarah left se bahar)

- Jab open hoti hai, JS `.show` class add karta hai jo `transform` ko `translateX(0)` kar deta hai — CSS `transition` se ye movement **smooth slide** dikhta hai (screen se andar slide ho kar aana)
- Modal ki tarah is mein bhi ek `.offcanvas-backdrop` create hota hai (background dim karne ke liye) aur body scroll disable ho jata hai jab tak offcanvas open hai

Modal vs Offcanvas: Modal center mein pop hoti hai (fade + scale/translateY), Offcanvas edge se slide hoti hai (translateX/Y) — dono ka backdrop mechanism similar hai.

37. Scrollspy

Scrollspy automatically navbar/nav links ko **highlight** karta hai based on ke user page par currently kahan scroll kar raha hai.

```
html

<body data-bs-spy="scroll" data-bs-target="#navbar-example" tabindex="0">

  <nav id="navbar-example" class="navbar">
    <a class="nav-link" href="#section1">Section 1</a>
    <a class="nav-link" href="#section2">Section 2</a>
  </nav>

  <div id="section1">...</div>
  <div id="section2">...</div>

</body>
```

Peeche ki CSS/mechanism:

- Ye pura component **CSS se nahi, JavaScript se** kaam karta hai — koi special "scrollspy CSS" nahi hoti

- Bootstrap JS scroll event ko listen karta hai aur continuously calculate karta hai ke viewport mein currently konsa section (`(id)` wala element) visible hai
 - Jab scroll position kisi particular section ke andar aati hai, JS us section ke corresponding nav link par `(.active)` class add kar deta hai (aur baqi links se `(.active)` hata deta hai) — yehi `(.active)` class normal `(.nav-link.active)` styling use karti hai jo already dusre nav components mein defined hoti hai
-

38. Placeholder (Skeleton Loading)

Placeholder content load hone se pehle ek **skeleton/loading preview** dikhata hai — jaise "content yahan ayega" ka gray box.

```
html

<p class="placeholder-glow">
  <span class="placeholder col-6"></span>
  <span class="placeholder col-4"></span>
</p>
```

Peeche ki CSS/mechanism:

- `(.placeholder)` element ko `(display: inline-block)`, gray `(background-color)`, aur `(border-radius)` deta hai — koi text nahi, sirf ek gray block, jiski width `(col-*)` grid classes (jaise `(col-6)`) se control hoti hai (proportional width simulate karne ke liye)
- `(.placeholder-glow)` parent par lagane se andar ke sab `(.placeholder)` elements par ek CSS `(@keyframes)` **opacity pulse animation** apply hoti hai — opacity continuously thora kam-zyada hoti rehti hai, jisse "glowing/breathing" effect banta hai
- **Alternative:** `(.placeholder-wave)` — glow ki bajaye ek **horizontal shimmer wave** effect deta hai (CSS gradient animation, jaisa progress-bar-animated mein tha, lekin placeholder ke context mein)

```
html
```

```
<p class="placeholder-wave">  
  <span class="placeholder col-12"></span>  
</p>
```

Use case: Jab tak actual API data load ho raha ho, user ko blank white space ki bajaye ek "content coming soon" ka visual hint milta hai — perceived loading speed better lagti hai.

Cheat Sheet

Component	Key Classes	Underlying Mechanism
Modal	<code>modal</code> , <code>modal-dialog</code> , <code>modal-content</code> , <code>modal-lg/sm</code> , <code>modal-dialog-centered</code> , <code>data-bs-toggle="modal"</code>	<code>display:none→block</code> + <code>.fade</code> / <code>.show</code> opacity+translateY transition; JS injects <code>.modal-backdrop</code> ; body scroll locked
Toast	<code>toast</code> , <code>toast-header</code> , <code>toast-body</code> , <code>toast-container</code> , <code>data-bs-autohide</code>	Same fade/show transition as modal, but no backdrop; JS <code>setTimeout</code> -style auto-hide
Tooltip	<code>data-bs-toggle="tooltip"</code> , <code>title</code> , <code>data-bs-placement</code>	Needs JS init; Popper.js positions it; fade/show transition
Popover	<code>data-bs-toggle="popover"</code> , <code>data-bs-content</code>	Same as tooltip + Popper.js, but has header+body and CSS-triangle arrow
Offcanvas	<code>offcanvas</code> , <code>offcanvas-start/end/top/bottom</code> , <code>data-bs-toggle="offcanvas"</code>	<code>transform: translateX/Y()</code> off-screen → <code>.show</code> brings to <code>translate(0)</code> ; backdrop + body scroll lock
Scrollspy	<code>data-bs-spy="scroll"</code> , <code>data-bs-target</code>	Pure JS scroll-position tracking; toggles <code>.active</code> on nav links
Placeholder	<code>placeholder</code> , <code>placeholder-glow</code> , <code>placeholder-wave</code> , <code>col-*</code>	Gray block via <code>background-color</code> ; <code>@keyframes</code> opacity pulse (glow) or gradient shimmer (wave)

Practice Exercises

1. Ek modal banao jisme header (title + close), body (form), aur footer (Cancel/Save buttons) hon — `modal-dialog-centered` use karo.
 2. Ek success toast banao jo 3 seconds ke baad automatically hide ho jaye, `toast-container` ke andar bottom-right corner mein.
 3. Ek button par tooltip lagao jo hover par "Click to save" dikhaye, aur JS se initialize karna mat bhoolna.
 4. Ek info icon par popover lagao jisme title "Info" aur body mein ek explanation ho.
 5. Ek `offcanvas-start` sidebar banao mobile navigation ke liye, 3-4 links ke sath.
 6. Ek page banao jisme 3 sections hon aur `scrollspy` navbar links ko current section ke hisab se highlight kare.
 7. Ek card ka skeleton loading state banao `placeholder-glow` use kar ke — jaise image, title, aur text placeholder.
-

Active Recall — Modal, Toast, Tooltip, Popover, Offcanvas, Scrollspy, Placeholder

| Khud se answer karne ki koshish karo, phir neeche scroll karke check karo.

Q1. Modal open hone par JS sirf `.modal` display change karta hai, ya kuch aur bhi karta hai (background ke liye)?

Q2. `modal-dialog-centered` kya CSS mechanism use karti hai modal ko vertically center karne ke liye?

Q3. Toast, Modal se mechanism mein kaise different hai (backdrop ke context mein)?

Q4. `data-bs-autohide` aur `data-bs-delay` ka toast mein kya role hai?

Q5. Tooltip aur Popover ko kaam karne ke liye extra kya step zaroori hai jo baqi components (jaise modal, collapse) ke liye nahi hota?

Q6. Tooltip ki positioning kis library se calculate hoti hai?

Q7. Tooltip aur Popover mein structural farq kya hai?

Q8. Offcanvas apni default (closed) state mein screen se "bahar" kaise rehta hai — konsi CSS property use hoti hai?

Q9. Offcanvas aur Modal ke opening mechanism mein basic farq kya hai (translate direction ke lehaz se)?

Q10. Scrollspy CSS-based hai ya JS-based? Ye kya calculate karta hai?

Q11. `placeholder-glow` aur `placeholder-wave` mein animation ka farq kya hai?

Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

A1. Nahi, sirf display change nahi karta — JS ek separate `.modal-backdrop` element bhi body mein inject karta hai (dark overlay background ke liye), aur `<body>` par `overflow: hidden` bhi laga deta hai taake background scroll na ho.

A2. `.modal-dialog-centered` `.modal-dialog` ko `display: flex` + `align-items: center` deta hai (sath `min-height` calculation), jisse modal box vertically viewport ke center mein align ho jata hai.

A3. Toast koi backdrop create nahi karta aur background ko block nahi karta — user baqi page ke sath interact kar sakta hai. Modal ek backdrop banata hai jo background ko dim aur non-interactive bana deta hai.

A4. `data-bs-autohide="true"` batata hai ke toast automatically hide ho jaye (default true), aur `data-bs-delay` milliseconds mein specify karta hai ke kitni der baad — JS internally `setTimeout` jaisi logic se ye implement karta hai.

A5. Inhe explicitly JavaScript se **initialize** karna parta hai (jaise `new bootstrap.Tooltip(element)`), jo dusre components (modal, collapse, offcanvas) ke liye zaroori nahi hota — wo `data-bs-*` attributes se automatically kaam kar jate hain.

A6. Popper.js library se — ye Bootstrap ke JS bundle (`bootstrap.bundle.min.js`) mein already included hoti hai aur calculate karti hai ke tooltip/popover viewport ke kis side best fit hoga.

A7. Tooltip sirf simple text show karta hai (sirf `title` attribute se), jabke Popover title (`.popover-header`) aur rich body content (`.popover-body`, `data-bs-content`) dono show kar sakta hai — zyada detailed info ke liye.

A8. `transform: translateX()` (ya top/bottom ke liye `translateY()`) se offcanvas apni closed state mein screen ke bahar rakha jata hai — jaise `offcanvas-start` ke liye `translateX(-100%)`.

A9. Modal fade + scale/translateY (center se pop-in) transition use karti hai. Offcanvas `translateX()` ya `translateY()` (edge se slide-in) use karta hai — dono ke transition ka direction/nature different hai, lekin backdrop mechanism similar hai.

A10. Scrollspy pure **JS-based** hai (koi special CSS nahi) — ye continuously scroll position track karta hai aur calculate karta hai ke viewport mein currently konsa section visible hai, phir corresponding nav link par `.active` class toggle karta hai.

A11. `placeholder-glow` ek opacity-based pulse animation deta hai (`@keyframes` se opacity kam-zyada hoti hai — breathing effect). `placeholder-wave` ek horizontal shimmer/gradient wave effect deta hai jo left se right move karta hai.

</details>